CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EC)

April/May 2018

EC243 Electronic Devices and Measurement

Date: 02.04.2018, Monday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) What is mobility? Derive equation for current density J and conductivity σ. **[07]**

Q - 2 (a) Explain the working of NPN transistor with necessary circuit diagram. **[08]**

(b) Describe construction and working of depletion mode metal oxide semiconductor field effect transistor (MOSFET) with necessary diagram. **[07]**

OR

Q - 2 (a) Draw and explain common emitter transistor configuration with its necessary input and output characteristics. **[08]**

(b) In a self-bias n- channel JFET, the operating point is to be set at I_D=1.5mA and V_{DS}=10V. The JFET parameters are I_{DSS} =5mA and $V_{GS(off)}$= - 2V. Find the values of R_S and R_D. Given that V_{DD}=20V. **[07]**

Q - 3 (a) Define shorted gate drain current. **[01]**

(b) Explain transistor direct coupled amplifier with special reference to frequency response, advantages, disadvantages and applications. **[06]**

(c) Define the terms Accuracy, errors, precision, resolution, sensitivity, instrument and probable error. **[06]**

OR

Q - 3 (a) Define shorted gate to source cutoff voltage. **[01]**

(b) Draw circuit diagram of RC coupled multi stage amplifier at high frequency and at low frequency. **[06]**

(c) (i) State three major categories of error. **[04]**

(ii) What is limiting errors or Guarantee errors? **[02]**

SECTION – II

Q - 4 (a) Which bridge is used for frequency measurement? Draw its circuit and derive the equation for frequency. Give its application and limitations. **[07]**

Q - 5 (a) Define the sensitivity and resolution of a digital meter. Explain with the help of a neat diagram the working of a DFM. **[08]**

(b) List out the applications of CRO and explain them in detail. **[07]**

OR

Q - 5 (a) What are the advantages of digital instruments over analog instruments? Explain the basic circuit for measurement of frequency showing Gate Control F/F. **[08]**

(b) Draw the structure of a Light Emitting Diode (LED) and explain its operation. **[07]**

Q - 6 (a) Write a note on basic sine wave generator. **[06]**

(b) Define Transducer. Give the classification of transducer. Which are the essential components of the LVDT? **[07]**

OR

Q - 6 (a) Explain fundamental block diagram of function generator. **[06]**

(b) Explain the important characteristics of thermistor with the help of necessary figures. **[07]**

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